

Practice Questions (Chapter 2 - Chapter 5)

CHAPTER 2

Question 1. An economy produces three goods: cars, computers, and oranges. Quantities and prices per unit for years 2012 and 2013 are as follows:

	2012		2013	
	Quantity	Price	Quantity	Price
Cars	10	\$2000	12	\$3000
Computers	4	\$1000	6	\$500
Oranges	1,000	\$1	1000	\$1

- 1) What is nominal GDP in 2012 and in 2013? By what percentage does nominal GDP change from 2012 to 2013?
- 2) Using the prices for 2012 as the set of common prices, what is real GDP in 2012 and in 2013? By what percentage does real GDP change from 2012 to 2013?
- 3) Using the prices for 2013 as the set of common prices, what is the real GDP in 2012 and in 2013? By what percentage does real GDP change from 2012 to 2013?
- 4) Why are two output growth rates constructed in 2) and 3) different? Which ones is correct? Explain your answer.

CHAPTER 3

Problem 1. Suppose that the economy is characterized by the following behavioral equations:

$$C = 160 + 0.6Y_D$$

$$I = 150$$

$$G = 150$$

$$T = 100$$

Solve for following variables.

- 1) Equilibrium GDP (Y)
- 2) Disposable income (Y_D)
- 3) Consumption spending (C)
- 4) Compute the total demand. Is it equal to production? Explain.

Assume that G is now equal to 110.

- 5) Solve for equilibrium output. Compute total demand. Is it equal to production? Explain.
- 6) Compute private plus public saving. Is the sum of private and public saving equal to investment? Explain.

CHAPTER 4

Problem 1. Consider a bond that promises to pay \$100 in one year.

- 1) What is the interest rate on the bond if its price today is \$75? \$85? \$95?
- 2) What is the relation between the price of the bond and the interest rate?
- 3) If the interest rate is 8%, what is the price of the bond today?

Problem 2. Suppose that money demand is given by

$$M^d = \$Y(0.25 - i)$$

where \$Y is \$100. Also, suppose that the supply of money is \$20.

- 1) What is the equilibrium interest rate?
- 2) If the Federal Reserve Bank wants to increase the equilibrium interest rate i by 10 percentage points from its value in 1), at what level should it set the supply of the money?

CHAPTER 5

Problem 1. Consider first the goods market model with constant investment that we saw in chapter 3. Consumption is given by

$$C = c_0 + c_1(Y - T)$$

and I , G , and T are given.

- 1) Solve for equilibrium output. What is the value of the multiplier for a change in autonomous spending?

Now let investment depend on both sales and the interest rate:

$$I = b_0 + b_1Y - b_2i$$

- 2) Solve for equilibrium output using the methods learned in Chapter 3. At a given interest rate, why is the effect of a change in autonomous spending bigger than what it was in part 1)? (Assume $c_1 + b_1 < 1$.)
- 3) Suppose the central bank chooses an interest rate of $\bar{i}$. Solve for equilibrium output at that interest rate.
- 4) Draw the equilibrium of this economy using an $IS - LM$ diagram.